

Formative Assessment 4

DSC1105 Exploratory Data Analysis

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GitHub Link:

<https://github.com/ChewyGnome/DSC1105>

Dataset

The dataset that will be used in this report contains chemical measurements of red and white wines. In the previous dataset, the objective was to predict wine quality, for this report the primary focus lies on identifying the difference between red and white wine types.

For this report the variables `total_sulfur_dioxide` and `density` were identified to be the primary predictors.

```
red_wine <- read.csv("https://archive.ics.uci.edu/ml/machine-learning-databases/wine-quality/winequality-red.csv")
white_wine <- read.csv("https://archive.ics.uci.edu/ml/machine-learning-databases/wine-quality/winequality-white.csv")

red_wine$type <- "red"
white_wine$type <- "white"

wine <- rbind(red_wine, white_wine)

names(wine) <- gsub("\\.", "_", names(wine))
```

Data Inspection

Structure of the Dataset

```
class(wine)
```

```
## [1] "data.frame"
```

First 6 observations of the Dataset

```
head(wine)
```

```
##   fixed_acidity volatile_acidity citric_acid residual_sugar chlorides
## 1           7.4           0.70           0.00           1.9      0.076
## 2           7.8           0.88           0.00           2.6      0.098
## 3           7.8           0.76           0.04           2.3      0.092
## 4          11.2           0.28           0.56           1.9      0.075
## 5           7.4           0.70           0.00           1.9      0.076
## 6           7.4           0.66           0.00           1.8      0.075
##   free_sulfur_dioxide total_sulfur_dioxide density    pH sulphates alcohol
## 1                  11                   34 0.9978 3.51      0.56     9.4
## 2                  25                   67 0.9968 3.20      0.68     9.8
## 3                  15                   54 0.9970 3.26      0.65     9.8
## 4                  17                   60 0.9980 3.16      0.58     9.8
## 5                  11                   34 0.9978 3.51      0.56     9.4
## 6                  13                   40 0.9978 3.51      0.56     9.4
##   quality type
## 1         5  red
## 2         5  red
## 3         5  red
## 4         6  red
## 5         5  red
## 6         5  red
```

Summary of the Dataset

```
summary(wine)
```

```
##   fixed_acidity    volatile_acidity    citric_acid    residual_sugar
##   Min.   : 3.800    Min.   :0.0800    Min.   :0.0000    Min.   : 0.600
##   1st Qu.: 6.400    1st Qu.:0.2300    1st Qu.:0.2500    1st Qu.: 1.800
##   Median : 7.000    Median :0.2900    Median :0.3100    Median : 3.000
##   Mean   : 7.215    Mean   :0.3397    Mean   :0.3186    Mean   : 5.443
##   3rd Qu.: 7.700    3rd Qu.:0.4000    3rd Qu.:0.3900    3rd Qu.: 8.100
##   Max.   :15.900    Max.   :1.5800    Max.   :1.6600    Max.   :65.800
##   chlorides    free_sulfur_dioxide    total_sulfur_dioxide    density
##   Min.   :0.00900    Min.   : 1.00    Min.   : 6.0    Min.   :0.9871
##   1st Qu.:0.03800    1st Qu.: 17.00    1st Qu.: 77.0    1st Qu.:0.9923
##   Median :0.04700    Median : 29.00    Median :118.0    Median :0.9949
##   Mean   :0.05603    Mean   : 30.53    Mean   :115.7    Mean   :0.9947
##   3rd Qu.:0.06500    3rd Qu.: 41.00    3rd Qu.:156.0    3rd Qu.:0.9970
##   Max.   :0.61100    Max.   :289.00    Max.   :440.0    Max.   :1.0390
##   pH    sulphates    alcohol    quality
##   Min.   :2.720    Min.   :0.2200    Min.   : 8.00    Min.   :3.000
##   1st Qu.:3.110    1st Qu.:0.4300    1st Qu.: 9.50    1st Qu.:5.000
##   Median :3.210    Median :0.5100    Median :10.30    Median :6.000
##   Mean   :3.219    Mean   :0.5313    Mean   :10.49    Mean   :5.818
##   3rd Qu.:3.320    3rd Qu.:0.6000    3rd Qu.:11.30    3rd Qu.:6.000
##   Max.   :4.010    Max.   :2.0000    Max.   :14.90    Max.   :9.000
##   type
```

```
## Length:6497
## Class :character
## Mode :character
##
##
##
```

Dataset structure

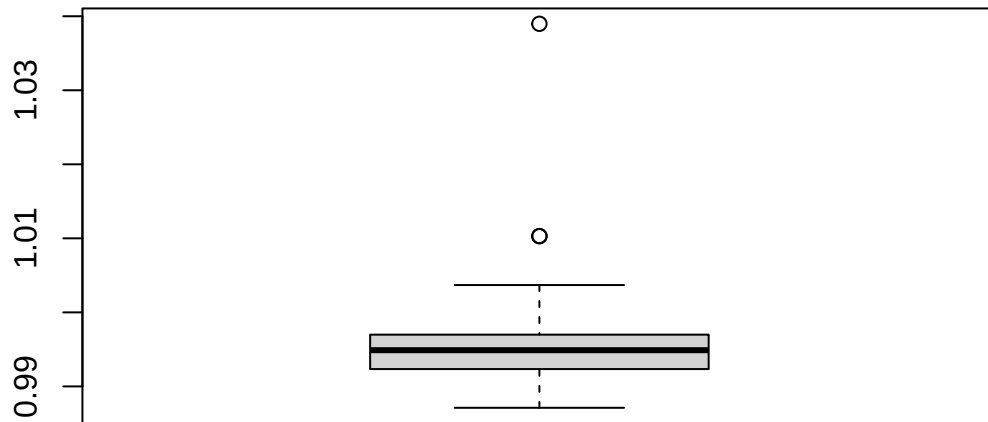
```
str(wine)
```

```
## 'data.frame': 6497 obs. of 13 variables:
## $ fixed_acidity : num 7.4 7.8 7.8 11.2 7.4 7.4 7.9 7.3 7.8 7.5 ...
## $ volatile_acidity : num 0.7 0.88 0.76 0.28 0.7 0.66 0.6 0.65 0.58 0.5 ...
## $ citric_acid : num 0 0 0.04 0.56 0 0 0.06 0 0.02 0.36 ...
## $ residual_sugar : num 1.9 2.6 2.3 1.9 1.9 1.8 1.6 1.2 2 6.1 ...
## $ chlorides : num 0.076 0.098 0.092 0.075 0.076 0.075 0.069 0.065 0.073 0.071 ...
## $ free_sulfur_dioxide : num 11 25 15 17 11 13 15 15 9 17 ...
## $ total_sulfur_dioxide: num 34 67 54 60 34 40 59 21 18 102 ...
## $ density : num 0.998 0.997 0.997 0.998 0.998 ...
## $ pH : num 3.51 3.2 3.26 3.16 3.51 3.51 3.3 3.39 3.36 3.35 ...
## $ sulphates : num 0.56 0.68 0.65 0.58 0.56 0.56 0.46 0.47 0.57 0.8 ...
## $ alcohol : num 9.4 9.8 9.8 9.8 9.4 9.4 9.4 10 9.5 10.5 ...
## $ quality : int 5 5 5 6 5 5 5 7 7 5 ...
## $ type : chr "red" "red" "red" "red" ...
```

Data Cleaning

Removing Outliers in Density

```
boxplot(wine$density)
```



```
Q1 <- quantile(wine$density, 0.25)
Q3 <- quantile(wine$density, 0.75)
IQR_val <- IQR(wine$density)

lower_bound <- Q1 - 1.5 * IQR_val
upper_bound <- Q3 + 1.5 * IQR_val

wine_clean <- wine %>%
  filter(density >= lower_bound & density <= upper_bound)
```

Exploratory Data Analysis

Variable Identification

Response variable: wine type (converted to binary)

Primary explanatory variables:

-total_sulfur_dioxide -density

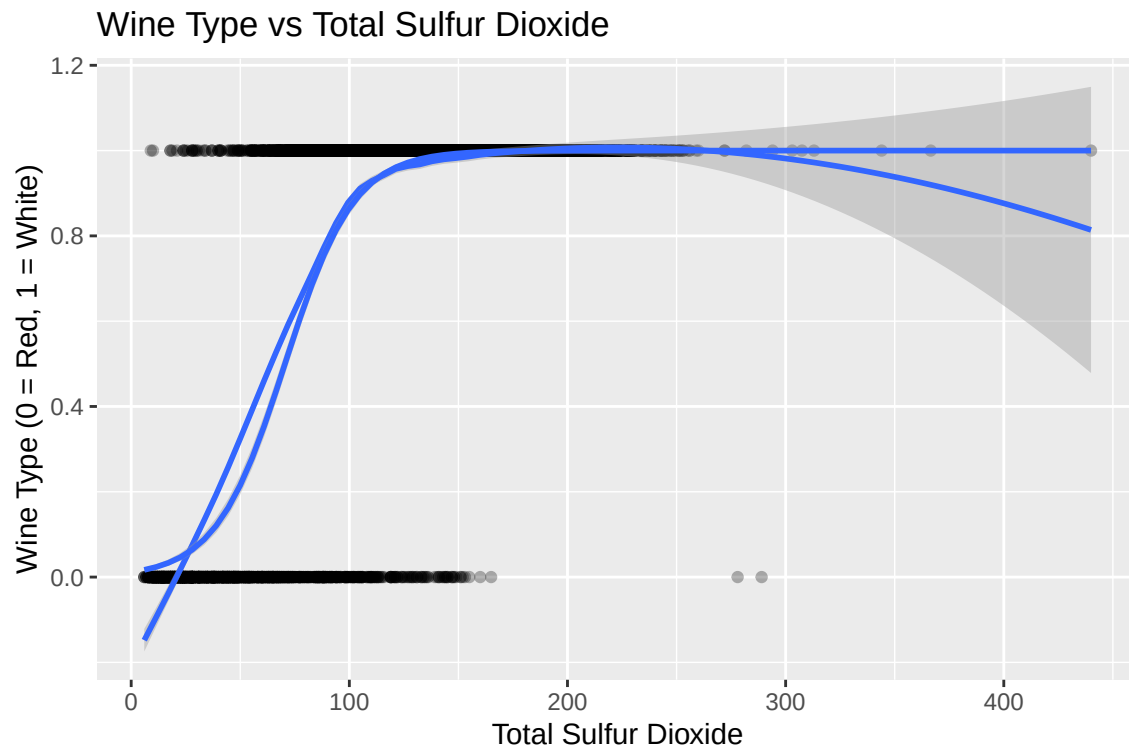
Convert Wine Type to Binary

```
wine_clean$type_binary <- ifelse(wine_clean$type == "white", 1, 0)
```

Scatterplot: Wine Type vs Total Sulfur Dioxide

```
ggplot(wine_clean, aes(x = total_sulfur_dioxide, y = type_binary)) +  
  geom_point(alpha = 0.3) +  
  geom_smooth(method = "loess") +  
  geom_smooth(method = "glm", method.args = list(family = "binomial")) +  
  labs(  
    title = "Wine Type vs Total Sulfur Dioxide",  
    x = "Total Sulfur Dioxide",  
    y = "Wine Type (0 = Red, 1 = White)"  
  )
```

```
## 'geom_smooth()' using formula = 'y ~ x'  
## 'geom_smooth()' using formula = 'y ~ x'
```



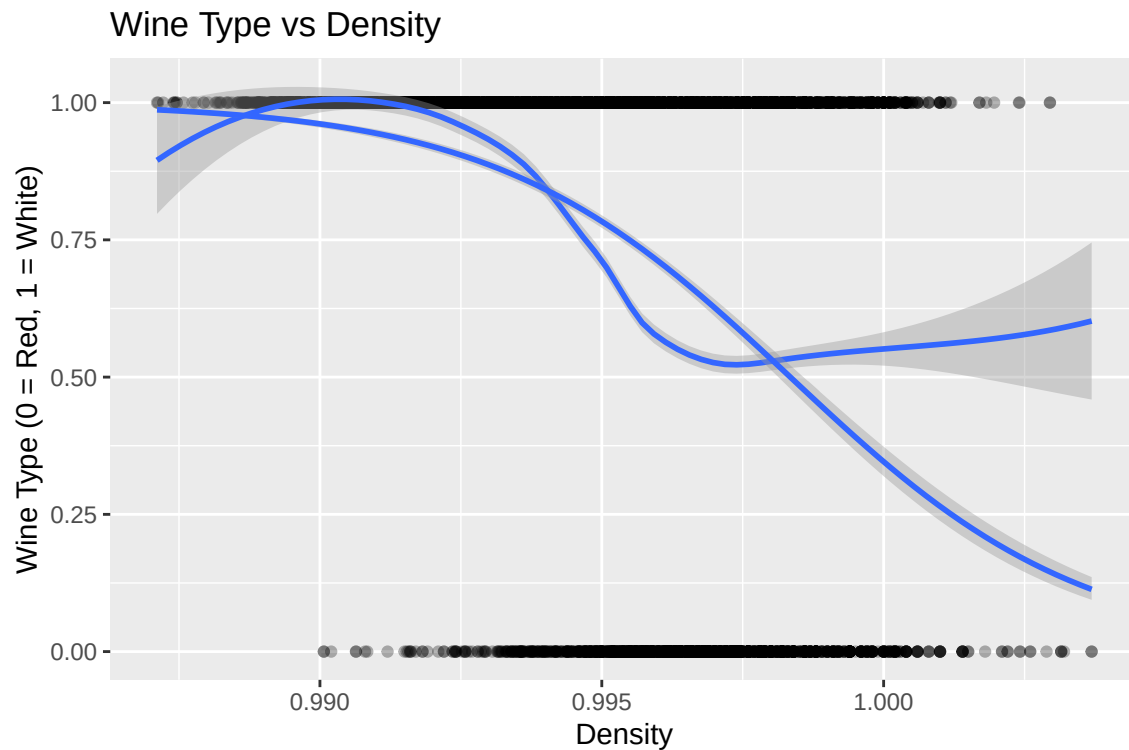
Interpretation: The LOESS smoother captures local trends in the data, while the logistic regression smoother gives a more objective interpretation. Differences between the curves indicate potential non-linearity in the relationship.

Scatterplot: Wine Type vs Density

```
ggplot(wine_clean, aes(x = density, y = type_binary)) +  
  geom_point(alpha = 0.3) +  
  geom_smooth(method = "loess") +  
  geom_smooth(method = "glm", method.args = list(family = "binomial")) +
```

```
labs(
  title = "Wine Type vs Density",
  x = "Density",
  y = "Wine Type (0 = Red, 1 = White)"
)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
## 'geom_smooth()' using formula = 'y ~ x'
```



Interpretation: The correlation between density and wine type shows a clear distinction, in this plot logistic regression highlights the classification boundaries with LOESS identifying local deviations.

Logistic Regression Models

Model 1: No Interaction

```
model1 <- glm(type_binary ~ density + total_sulfur_dioxide,
  data = wine_clean,
  family = "binomial")

summary(model1)
```

```
##
## Call:
## glm(formula = type_binary ~ density + total_sulfur_dioxide, family = "binomial",
```

```
##      data = wine_clean)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      802.43997    32.60225   24.61  <2e-16 ***
## density          -811.91260    32.86373  -24.70  <2e-16 ***
## total_sulfur_dioxide  0.07291     0.00230   31.70  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 7249.3  on 6493  degrees of freedom
## Residual deviance: 1607.5  on 6491  degrees of freedom
## AIC: 1613.5
##
## Number of Fisher Scoring iterations: 7
```

Model 2: With Interaction

```
model2 <- glm(type_binary ~ density * total_sulfur_dioxide,
              data = wine_clean,
              family = "binomial")
```

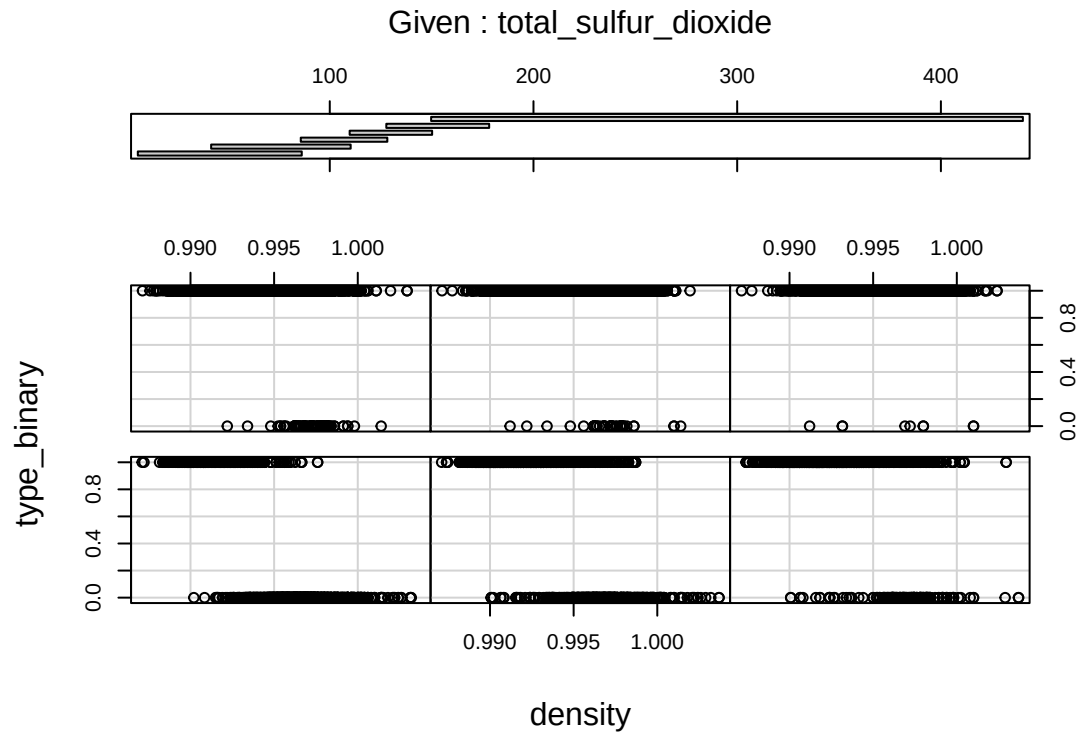
```
summary(model2)
```

```
##
## Call:
## glm(formula = type_binary ~ density * total_sulfur_dioxide, family = "binomial",
##      data = wine_clean)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      1646.544    108.283   15.206  <2e-16 ***
## density          -1660.289    108.905  -15.245  <2e-16 ***
## total_sulfur_dioxide  -8.578      1.004   -8.545  <2e-16 ***
## density:total_sulfur_dioxide  8.691      1.009   8.612  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 7249.3  on 6493  degrees of freedom
## Residual deviance: 1515.4  on 6490  degrees of freedom
## AIC: 1523.4
##
## Number of Fisher Scoring iterations: 8
```

Coplots

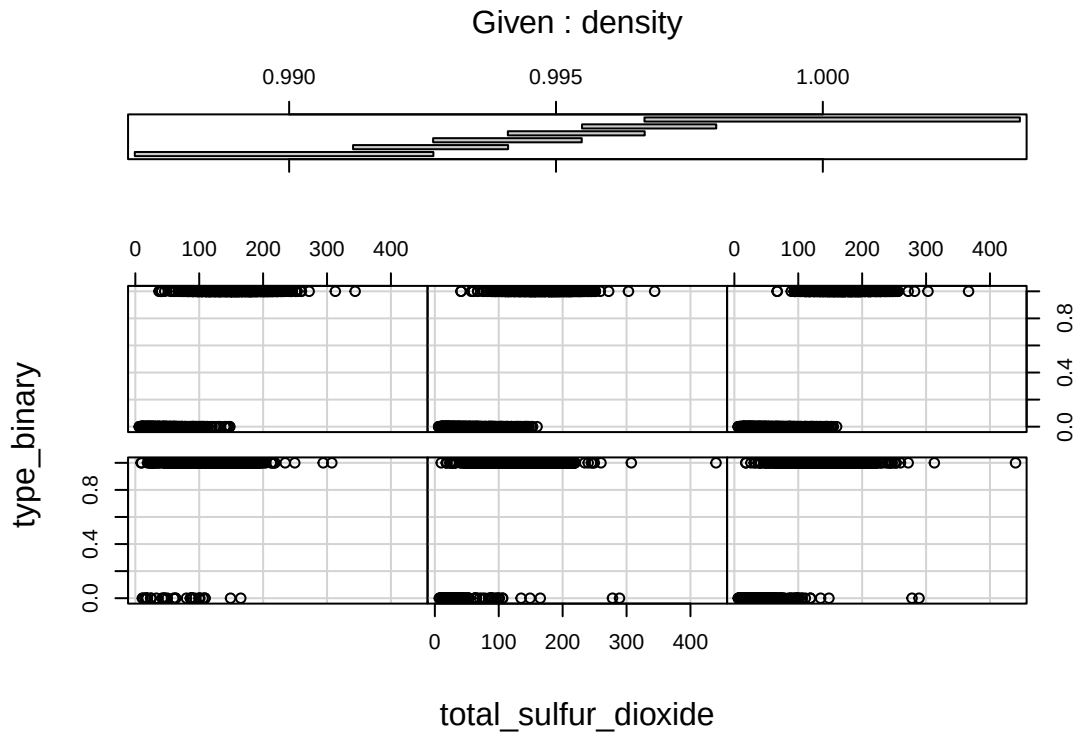
Density conditioned on Total Sulfur Dioxide

```
coplot(type_binary ~ density | total_sulfur_dioxide, data = wine_clean)
```



Total Sulfur Dioxide conditioned on Density

```
coplot(type_binary ~ total_sulfur_dioxide | density, data = wine_clean)
```

Model Comparison Interpretation

This interaction model allows the effect of density on wine type to vary depending on the level of total sulfur dioxide. This provides a more flexible model compared to the additive model without interaction.

Reflection

In this report, the relationship between wine type and key chemical properties were analyzed to see if there is significant correlation between the two. Variables within chemical properties such as density and total sulfur dioxide were observed through the implementation of visualization and modeling technique. From the results of the models it was clear that both played a significant roles in distinguishing between red and white wines.

Additionally, the use of LOESS and logistic regression smoothers highlights both the local and global patterns in the data. Logistic regression models further quantified these relationships, showing how predictor variables influence the probability of a wine being classified as red or white.

Concluding this activity, it was shown how important the implementation of visualization and statistical modeling in understanding complex datasets.